

SIMATS ENGINEERING
ASSESSMENT TOOL 2: LABORATORY PERFORMANCE

COURSE NAME:	Programming in Java	COURSE CODE:	CSA0915- SLOT B
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ASSESSMENT TASKS:

3. Develop an AWT-based Student Registration Form using: Label

TextField

CheckboxGroup

Choice

Button

Frame- Arrange the components using an appropriate layout manager. "

Test Cases:

tc1_input- Student ID=S101, Name=Ravi Kumar, Gender=Male,

Department=Information Technology

tc1_output- Student Registered Successfully

tc2_input- Student ID=S102, Name=Priya, Gender=Female, Department=Computer
Science

tc2_output- Registration Successful and student details displayed

tc3_input- Student Name is empty

tc3_output- Student Name is required

CODE:

A. JAVA CODE

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class StudentRegistration extends Frame implements ActionListener {
```

Label idLabel, nameLabel, genderLabel, deptLabel;

TextField idField, nameField;

CheckboxGroup genderGroup;

Checkbox male, female;

Choice department;

Button registerButton, clearButton;

TextArea resultArea;

public StudentRegistration() {

setTitle("Student Registration Form");

setSize(600, 450);

setLayout(new BorderLayout(10, 10));

Label title = new Label("STUDENT REGISTRATION FORM", Label.CENTER);

title.setFont(new Font("Arial", Font.BOLD, 20));

add(title, BorderLayout.NORTH);

Panel formPanel = new Panel();

formPanel.setLayout(new GridLayout(4, 2, 10, 10));

idLabel = new Label("Student ID:");

idField = new TextField();

nameLabel = new Label("Student Name:");

nameField = new TextField();

```
genderLabel = new Label("Gender:");
```

```
genderGroup = new CheckboxGroup();
```

```
male = new Checkbox("Male", genderGroup, false);
```

```
female = new Checkbox("Female", genderGroup, false);
```

```
Panel genderPanel = new Panel(new FlowLayout(FlowLayout.LEFT));
```

```
genderPanel.add(male);
```

```
genderPanel.add(female);
```

```
deptLabel = new Label("Department:");
```

```
department = new Choice();
```

```
department.add("Select Department");
```

```
department.add("Information Technology");
```

```
department.add("Computer Science");
```

```
department.add("Artificial Intelligence");
```

```
department.add("Electronics and Communication");
```

```
department.add("Mechanical Engineering");
```

```
formPanel.add(idLabel);
```

```
formPanel.add(idField);
```

```
formPanel.add(nameLabel);
```

```
formPanel.add(nameField);
```

```
formPanel.add(genderLabel);
```

```
formPanel.add(genderPanel);
```

```
formPanel.add(deptLabel);
```

```
formPanel.add(department);
```

```
add(formPanel, BorderLayout.CENTER);
```

```
Panel bottomPanel = new Panel(new BorderLayout());
```

```
Panel buttonPanel = new Panel();
```

```
registerButton = new Button("Register");
```

```
clearButton = new Button("Clear");
```

```
registerButton.addActionListener(this);
```

```
clearButton.addActionListener(this);
```

```
buttonPanel.add(registerButton);
```

```
buttonPanel.add(clearButton);
```

```
bottomPanel.add(buttonPanel, BorderLayout.NORTH);
```

```
resultArea = new TextArea("", 8, 50);
```

```
resultArea.setEditable(false);
```

```
bottomPanel.add(resultArea, BorderLayout.CENTER);

add(bottomPanel, BorderLayout.SOUTH);

addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        System.exit(0);
    }
});

setVisible(true);
}

public void actionPerformed(ActionEvent e) {

    if (e.getSource() == registerButton) {
        registerStudent();
    }

    if (e.getSource() == clearButton) {
        clearFields();
    }
}

private void registerStudent() {
```

```
String studentId = idField.getText().trim();  
String studentName = nameField.getText().trim();
```

```
if (studentId.isEmpty()) {  
    resultArea.setText("Student ID is required");  
    return;  
}
```

```
if (studentName.isEmpty()) {  
    resultArea.setText("Student Name is required");  
    return;  
}
```

```
Checkbox selectedGender = genderGroup.getSelectedCheckbox();
```

```
if (selectedGender == null) {  
    resultArea.setText("Gender is required");  
    return;  
}
```

```
if (department.getSelectedIndex() == 0) {  
    resultArea.setText("Department is required");  
    return;  
}
```

```
String gender = selectedGender.getLabel();  
String dept = department.getSelectedItemId();
```

```
resultArea.setText(  
    "Student Registered Successfully\n\n" +  
    "Student ID : " + studentId + "\n" +  
    "Name      : " + studentName + "\n" +  
    "Gender    : " + gender + "\n" +  
    "Department : " + dept  
);  
}
```

```
private void clearFields() {  
  
    idField.setText("");  
    nameField.setText("");  
    genderGroup.setSelectedCheckbox(null);  
    department.select(0);  
    resultArea.setText("");  
}
```

```
public static void main(String[] args) {  
    new StudentRegistration();  
}  
}
```

OUTPUT:

 Student Registration Form—□×

STUDENT REGISTRATION FORM

Student ID:

101

Student Name:

JESHWANT

Gender:

☒ Male ☐ Female

Department:

Information Technology

Register

Clear

Student Registered Successfully

Student ID : 101
Name : JESHWANT
Gender : Male
Department : Information Technology